UNIT 6 — SIMPLE MACHINES

Revision notes based on the supplied textbook pages. Includes definitions, principles, lever classes, uses and examples.

1. WHAT IS A MACHINE?

• A machine is a device that helps us to do work more easily.

• Machines can help us in four main ways: multiply force, apply force in a convenient direction, save time, and assist in doing unsafe or dangerous jobs.

2. TYPES OF SIMPLE MACHINES

The five simple machines listed in the textbook are: lever, pulley, inclined plane, screw and wheel.

Simple machine	Basic idea / use	Examples
Lever	A rigid bar that turns about a fixed point called the fulcrum.	See-saw, scissors, bottle opener, wheelbarrow, forceps
Pulley	A grooved wheel over which a rope or chain passes; helps lift loads in a convenient direction.	Drawing water from a well, cranes, lifts
Inclined plane	A sloping surface used to raise or lower a load with less effort over a longer distance.	Ramp, staircase, sloping road
Screw	A modified inclined plane wrapped around a cylinder; converts turning motion into forward motion.	Screw jack, carpenter's screw, bolts
Wheel	A circular device that rolls and helps movement with less friction.	Wheels of vehicles, rollers, ball bearings

3. LEVERS — IMPORTANT TERMS

• A lever is a rigid rod that is free to turn about a fixed point called the fulcrum.

• Load: the object or resistance that is to be moved or lifted.

• Effort: the force applied to move or lift the load.

• Effort arm: distance between the fulcrum and the point where effort acts.

• Load arm: distance between the fulcrum and the load.

4. PRINCIPLE OF LEVERS

For a balanced lever: Effort × Effort arm = Load × Load arm

Mechanical advantage (M.A.) of a lever: M.A. = Load / Effort = Effort arm / Load arm (for an ideal balanced lever).

• A longer effort arm compared with the load arm can reduce the effort needed to lift a load.

5. THREE CLASSES OF LEVERS

Class	Relative positions	Examples
First-order (first class)	Fulcrum lies between the load and effort.	See-saw, scissors, beam balance, tin shears, crowbar, pliers
Second-order (second class)	Load lies between the fulcrum and effort.	Wheelbarrow, nutcracker, bottle opener, nail cutter
Third-order (third class)	Effort lies between the fulcrum and load.	Forceps, human forearm, spade, fishing rod

Memory aid

First class: F in the middle. Second class: L in the middle. Third class: E in the middle. (F = fulcrum, L = load, E = effort.)

6. PULLEY

• A pulley is a simple machine that helps lift heavy loads by applying effort in a suitable or convenient direction.

• A fixed pulley can change the direction of effort. For example, pulling down on a rope can raise a bucket from a well.

• Pulleys are used in wells, cranes, lifts and some vehicle-lifting arrangements.

7. INCLINED PLANE

• An inclined plane is a sloping surface used to raise a heavy load more easily than lifting it vertically.

• The load moves through a longer distance along the slope, but a smaller effort is generally needed.

• A steeper slope requires greater force than a gentler slope for the same load and height.

• Examples: ramps, staircases and sloping roads. A knife, axe and chisel are wedges formed by joining inclined planes.

8. SCREW

• A screw is a modified inclined plane wrapped around a cylinder, producing a helical thread.

• Turning a screw converts circular motion into forward or linear motion.

• A screw jack can be used to lift heavy objects such as cars. A screw's thread grips wood firmly, making it harder to pull out than an ordinary nail.

9. WHEEL AND FRICTION

• A wheel helps objects move with less effort. Rolling friction is generally less than sliding friction.

• Wheels and rollers reduce the force needed to move heavy objects.

• Ball bearings are used in moving parts of machines to reduce friction.

10. QUICK COMPARISON AND EXAM FACTS

Question / clue	Answer
Fulcrum between load and effort	First-class lever
Load between fulcrum and effort	Second-class lever
Effort between fulcrum and load	Third-class lever
Bottle opener / wheelbarrow	Second-class lever
See-saw / scissors / beam balance	First-class lever
Forceps / forearm / fishing rod	Third-class lever
Drawing water from a well	Pulley
Ramp / staircase	Inclined plane
Thread wrapped around a cylinder	Screw
Rolling friction compared with sliding friction	Usually smaller

11. COMMON MCQ TRAPS

• A lever is not defined by the object's appearance; identify the positions of fulcrum, load and effort.

• A wheelbarrow is a second-class lever because its load is between the wheel (fulcrum) and the handles (effort).

• A see-saw is a first-class lever because its fulcrum is between the two ends.

• A forearm lifting a weight is a third-class lever because the effort from the biceps acts between the elbow (fulcrum) and the load in the hand.

• An inclined plane reduces the effort by increasing the distance over which the effort acts; it does not remove the work required.

Exam tip: For lever questions, mark F, L and E first. Their relative positions immediately identify the class.